

AMC1311x High-Impedance, 2-V Input, Reinforced Isolated Amplifiers

1 Features

- 2-V, high-impedance input voltage range optimized for isolated voltage measurement
- Fixed gain: 1
- Low DC errors:
 - AMC1311:
 - Offset error: ± 9.9 mV (maximum)
 - Offset drift: ± 20 $\mu\text{V}/^\circ\text{C}$ (typical)
 - Gain error: $\pm 1\%$ (maximum)
 - Gain drift: ± 30 ppm/ $^\circ\text{C}$ (typical)
 - AMC1311B:
 - Offset error: ± 1.5 mV (maximum)
 - Offset drift: ± 10 $\mu\text{V}/^\circ\text{C}$ (maximum)
 - Gain error: $\pm 0.2\%$ (maximum)
 - Gain drift: ± 40 ppm/ $^\circ\text{C}$ (maximum)
 - Nonlinearity: 0.04% (maximum)
- 3.3-V operation on high-side (AMC1311B)
- High CMTI: 100 kV/ μs (minimum) (AMC1311B)
- Missing high-side supply indication
- Safety-related certifications:
 - 7000- V_{PK} reinforced isolation per DIN EN IEC 60747-17 (VDE 0884-17)
 - 5000- V_{RMS} isolation for 1 minute per UL1577
 - Fully specified over the extended industrial temperature range: -40°C to $+125^\circ\text{C}$

2 Applications

- Isolated voltage sensing in:
 - Motor drives
 - Frequency inverters
 - Uninterruptible power supplies

3 Description

The AMC1311 is a precision, isolated amplifier with an output separated from the input circuitry by a capacitive isolation barrier that is highly resistant to magnetic interference. This barrier is certified to provide reinforced galvanic isolation of up to 5 kV_{RMS} according to DIN EN IEC 60747-17 (VDE 0884-17) and UL1577 and supports a working voltage of up to 1500 V_{RMS}.

The isolation barrier separates parts of the system that operate on different common-mode voltage levels and protects the low-voltage side from voltages that can cause electrical damage or be harmful to an operator.

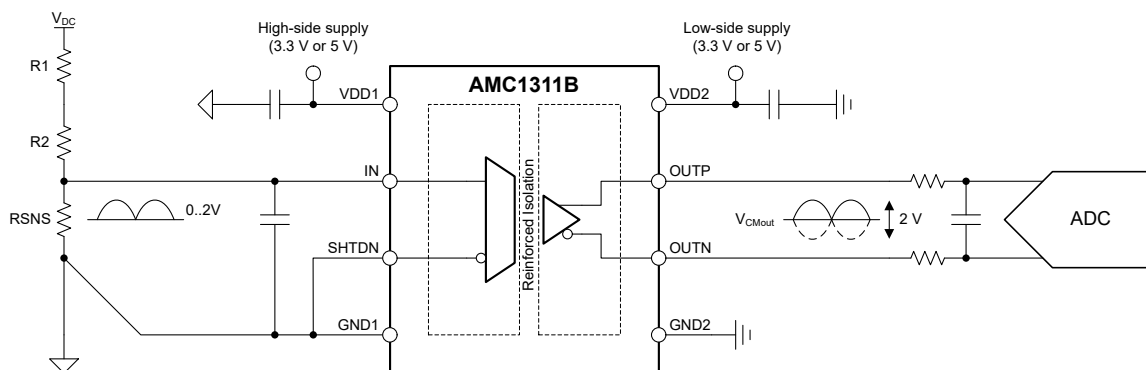
The high-impedance input of the AMC1311 is optimized for connection to high-impedance resistive dividers or any other high-impedance voltage signal source. The excellent DC accuracy and low temperature drift support accurate, isolated voltage sensing and control in closed-loop systems. The integrated missing high-side supply voltage detection feature simplifies system-level design and diagnostics.

The AMC1311 is offered with two performance grade options: the AMC1311B is specified over the extended industrial temperature range of -55°C to $+125^\circ\text{C}$, and the AMC1311 is specified for operation at -40°C to $+125^\circ\text{C}$.

Device Information⁽¹⁾

PART NUMBER	PACKAGE	BODY SIZE (NOM)
AMC1311	SOIC (8)	5.85 mm × 7.50 mm
AMC1311B		

- (1) For all available packages, see the orderable addendum at the end of the data sheet.



Simplified Schematic



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4 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from Revision B (May 2020) to Revision C (June 2022)	Page
• Changed isolation standard from DIN VDE V 0884-11 (VDE V 0884-11) to DIN EN IEC 60747-17 (VDE 0884-17) and updated the <i>Insulation Specifications</i> and <i>Safety-Related Certifications</i> tables accordingly.....	1
• Changed <i>Features</i> section.....	1
• Changed pin names: VIN to IN, VOUTP to OUTP, and VOUTN to OUTN.....	5
• Merged V _{OS} specs for 4.5V ≤ VDD1 ≤ 5.5 V and 3.0 V ≤ VDD1 ≤ 5.5 V ranges (AMC1311B only).....	10
• Changed VDD1 DC PSRR from –65 dB (typical) to –80 dB (typical).....	10
• Changed CMTI from 75 kV/μs (minimum), 140 kV/μs (typical) to 100 kV/μs (minimum), 150kV/μs (typical) (AMC1311B only).....	10
• Changed VDD1 _{UV} (VDD1 falling) from 1.75 V / 2.53 V / 2.7 V to 2.4 V / 2.6 V / 2.8 V (minimum / typical / maximum).....	10
• Changed <i>Rise, Fall, and Delay Time Definition</i> timing diagram.....	12
• Changed <i>Reinforced Isolation Capacitor Lifetime Projection</i> figure	13
• Changed functional block diagram.....	20
• Deleted <i>Fail-Safe Output</i> section, added <i>Analog Output</i> section.....	22
• Changed <i>Typical Application</i> section and subsections.....	23
• Changed <i>What To Do and What Not To Do</i> section.....	26
• Changed <i>Layout</i> section.....	28

Changes from Revision A (June 2018) to Revision B (May 2020)

Page

• Changed AMC1311B offset drift from $\pm 15 \mu\text{V}/^\circ\text{C}$ (max) to $10 \mu\text{V}/^\circ\text{C}$ (max) in <i>Features</i> section.....	1
• Changed AMC1311B gain error from $\pm 0.3\%$ (max) to $\pm 0.2\%$ (max) and changed AMC1311B gain drift from $\pm 45 \text{ ppm}/^\circ\text{C}$ (max) to $\pm 40 \text{ ppm}/^\circ\text{C}$ (max) in <i>Features</i> section.....	1
• Changed IEC 60950-1 and IEC60065 to IEC 62368-1	1
• Changed AMC1311B values for TCV_{OS} , E_{G} , and TCE_{G} in <i>Device Comparison Table</i>	4
• Changed AMC1311B values for TCV_{OS} , E_{G} , and TCE_{G} in <i>Device Comparison Table</i>	6
• Added ESD classification levels to <i>ESD Ratings</i> table.....	6
• Changed CLR and CPG values from 9 mm to 8.5 mm.....	6
• Changed <i>Insulation Specifications</i> table per ISO standard.....	6
• Changed <i>Safety-Related Certification</i> table per ISO standard.....	6
• Changed <i>Safety Limiting Values</i> description as per ISO standard.....	6
• Changed TCV_{OS} parameter minimum value from $-15 \mu\text{V}/^\circ\text{C}$ to $-10 \mu\text{V}/^\circ\text{C}$ and maximum value from $15 \mu\text{V}/^\circ\text{C}$ to $10 \mu\text{V}/^\circ\text{C}$ for the AMC1311B in the <i>Electrical Characteristics</i> table.....	6
• Changed E_{G} parameter minimum value from -0.3% to -0.2% and maximum value from 0.3% to 0.2% for the AMC1311B in the <i>Electrical Characteristics</i> table.....	6
• Changed TCE_{G} parameter minimum value from $-45 \text{ ppm}/^\circ\text{C}$ to $-40 \text{ ppm}/^\circ\text{C}$ and maximum value from $45 \text{ ppm}/^\circ\text{C}$ to $40 \text{ ppm}/^\circ\text{C}$ for the AMC1311B in the <i>Electrical Characteristics</i> table.....	6
• Changed <i>Step Response of the AMC1311</i> figure.....	26

5 Device Comparison Table

PARAMETER		AMC1311B	AMC1311
High-side supply voltage, VDD1		3.0 V to 5.5 V	4.5 V to 5.5 V
Specified ambient temperature, T _A		–55°C to +125°C	–40°C to +125°C
Input offset voltage, V _{OS}	4.5 V ≤ VDD1 ≤ 5.5 V	±1.5 mV	±9.9 mV
	3.0 V ≤ VDD1 ≤ 5.5 V	±2.5 mV	Not applicable
Input offset drift, TCV _{OS}		±3 μV/°C (typ), ±10 μV/°C (max)	±20 μV/°C (typ)
Gain error, E _G		±0.2%	±1%
Gain error drift, TCE _G		±5 ppm/°C (typ), ±40 ppm/°C (max)	±30 ppm/°C (typ)
Common-mode transient immunity, CMTI		100 kV/μs (min)	15 kV/μs (min)

6 Pin Configuration and Functions

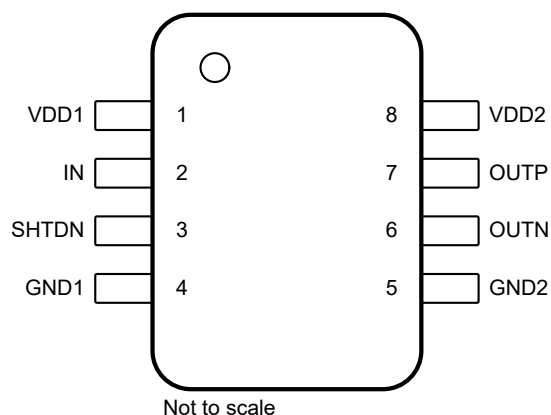


Figure 6-1. DWV Package, 8-Pin SOIC (Top View)

Table 6-1. Pin Functions

PIN		TYPE	DESCRIPTION
NO.	NAME		
1	VDD1	High-side power	High-side power supply ⁽¹⁾
2	IN	Analog input	Analog input
3	SHTDN	Digital input	Shutdown input, active high, with internal pullup resistor (typical value: 100 kΩ)
4	GND1	High-side ground	High-side analog ground
5	GND2	Low-side ground	Low-side analog ground
6	OUTN	Analog output	Inverting analog output
7	OUTP	Analog output	Noninverting analog output
8	VDD2	Low-side power	Low-side power supply ⁽¹⁾

(1) See the [Power Supply Recommendations](#) section for power-supply decoupling recommendations.

7 Specifications

7.1 Absolute Maximum Ratings

see⁽¹⁾

		MIN	MAX	UNIT
Power-supply voltage	High-side VDD1 to GND1	−0.3	6.5	V
	Low-side VDD2 to GND2	−0.3	6.5	
Input voltage	IN	GND1 − 6	VDD1 + 0.5	V
	SHTDN	GND1 − 0.5	VDD1 + 0.5	
Output voltage	OUTP, OUTN	GND2 − 0.5	VDD2 + 0.5	V
Input current	Continuous, any pin except power-supply pins	−10	10	mA
Temperature	Junction, T _J		150	°C
	Storage, T _{stg}	−65	150	

- (1) Operation outside the *Absolute Maximum Ratings* may cause permanent device damage. *Absolute Maximum Ratings* do not imply functional operation of the device at these or any other conditions beyond those listed under *Recommended Operating Conditions*. If used outside the Recommended Operating Conditions but within the Absolute Maximum Ratings, the device may not be fully functional, and this may affect device reliability, functionality, performance, and shorten the device lifetime.

7.2 ESD Ratings

			VALUE	UNIT
V _(ESD)	Electrostatic discharge	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001 ⁽¹⁾	±2000	V
		Charged-device model (CDM), per ANSI/ESDA/JEDEC JS-002 ⁽²⁾	±1000	

- (1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.
 (2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

7.3 Recommended Operating Conditions

over operating ambient temperature range (unless otherwise noted)

			MIN	NOM	MAX	UNIT
POWER SUPPLY						
	High-side power supply	VDD1 to GND1, AMC1311	4.5	5	5.5	V
		VDD1 to GND1, AMC1311B	3	5	5.5	
	Low-side power supply	VDD2 to GND2	3	3.3	5.5	V
ANALOG INPUT						
V _{Clipping}	Input voltage before clipping output	IN to GND1		2.516		V
V _{FSR}	Specified linear full-scale voltage	IN to GND1	−0.1		2	V
ANALOG OUTPUT						
C _{LOAD}	Capacitive load	On OUTP or OUTN to GND2			500	pF
		OUTP to OUTN			250	
R _{LOAD}	Resistive load	On OUTP or OUTN to GND2		10	1	kΩ
DIGITAL INPUT						
	Input voltage	SHTDN to GND1	0		VDD1	V
TEMPERATURE RANGE						
T _A	Specified ambient temperature	AMC1311	−40		125	°C
		AMC1311B	−55		125	

7.4 Thermal Information

THERMAL METRIC ⁽¹⁾		DWV (SOIC)	UNIT
		8 PINS	
R _{θJA}	Junction-to-ambient thermal resistance	84.6	°C/W
R _{θJC(top)}	Junction-to-case (top) thermal resistance	28.3	°C/W
R _{θJB}	Junction-to-board thermal resistance	41.1	°C/W
Ψ _{JT}	Junction-to-top characterization parameter	4.9	°C/W
Ψ _{JB}	Junction-to-board characterization parameter	39.1	°C/W
R _{θJC(bot)}	Junction-to-case (bottom) thermal resistance	n/a	°C/W

(1) For more information about traditional and new thermal metrics, see the [Semiconductor and IC Package Thermal Metrics](#) application report.

7.5 Power Ratings

PARAMETER		TEST CONDITIONS	VALUE	UNIT
P _D	Maximum power dissipation (both sides)	VDD1 = VDD2 = 5.5 V	98	mW
		VDD1 = VDD2 = 3.6V, AMC1311B only	56	
P _{D1}	Maximum power dissipation (high-side)	VDD1 = 5.5 V	53	mW
		VDD1 = 3.6 V, AMC1311B only	30	
P _{D2}	Maximum power dissipation (low-side)	VDD2 = 5.5 V	45	mW
		VDD2 = 3.6 V, AMC1311B only	26	

7.6 Insulation Specifications

over operating ambient temperature range (unless otherwise noted)

PARAMETER		TEST CONDITIONS	VALUE	UNIT
GENERAL				
CLR	External clearance ⁽¹⁾	Shortest pin-to-pin distance through air	≥ 8.5	mm
CPG	External creepage ⁽¹⁾	Shortest pin-to-pin distance across the package surface	≥ 8.5	mm
DTI	Distance through insulation	Minimum internal gap (internal clearance) of the double insulation	≥ 0.021	mm
CTI	Comparative tracking index	DIN EN 60112 (VDE 0303-11); IEC 60112	≥ 600	V
	Material group	According to IEC 60664-1	I	
	Overvoltage category per IEC 60664-1	Rated mains voltage ≤ 600 V _{RMS}	I-IV	
		Rated mains voltage ≤ 1000 V _{RMS}	I-III	
DIN EN IEC 60747-17 (VDE 0884-17) ⁽²⁾				
V _{IORM}	Maximum repetitive peak isolation voltage	At AC voltage	2120	V _{PK}
V _{IOWM}	Maximum-rated isolation working voltage	At AC voltage (sine wave)	1500	V _{RMS}
		At DC voltage	2120	V _{DC}
V _{IOTM}	Maximum transient isolation voltage	V _{TEST} = V _{IOTM} , t = 60 s (qualification test)	7000	V _{PK}
		V _{TEST} = 1.2 × V _{IOTM} , t = 1 s (100% production test)	8400	
V _{IMP}	Maximum impulse voltage ⁽³⁾	Tested in air, 1.2/50-μs waveform per IEC 62368-1	9800	V _{PK}
V _{IOSM}	Maximum surge isolation voltage ⁽⁴⁾	Tested in oil (qualification test), 1.2/50-μs waveform per IEC 62368-1	12800	V _{PK}
q _{pd}	Apparent charge ⁽⁵⁾	Method a, after input/output safety test subgroups 2 and 3, V _{ini} = V _{IOTM} , t _{ini} = 60 s, V _{pd(m)} = 1.2 × V _{IORM} , t _m = 10 s	≤ 5	pC
		Method a, after environmental tests subgroup 1, V _{ini} = V _{IOTM} , t _{ini} = 60 s, V _{pd(m)} = 1.6 × V _{IORM} , t _m = 10 s	≤ 5	
		Method b1, at routine test (100% production) and preconditioning (type test), V _{ini} = V _{IOTM} , t _{ini} = 1 s, V _{pd(m)} = 1.875 × V _{IORM} , t _m = 1 s	≤ 5	
C _{IO}	Barrier capacitance, input to output ⁽⁶⁾	V _{IO} = 0.5 V _{PP} at 1 MHz	~1.5	pF
R _{IO}	Insulation resistance, input to output ⁽⁶⁾	V _{IO} = 500 V at T _A = 25°C	> 10 ¹²	Ω
		V _{IO} = 500 V at 100°C ≤ T _A ≤ 125°C	> 10 ¹¹	
		V _{IO} = 500 V at T _S = 150°C	> 10 ⁹	
	Pollution degree		2	
	Climatic category		55/125/21	
UL1577				
V _{ISO}	Withstand isolation voltage	V _{TEST} = V _{ISO} = 5000 V _{RMS} , t = 60 s (qualification), V _{TEST} = 1.2 × V _{ISO} = 6000 V _{RMS} , t = 1 s (100% production test)	5000	V _{RMS}

- (1) Apply creepage and clearance requirements according to the specific equipment isolation standards of an application. Care must be taken to maintain the creepage and clearance distance of a board design to ensure that the mounting pads of the isolator on the printed circuit board (PCB) do not reduce this distance. Creepage and clearance on a PCB become equal in certain cases. Techniques such as inserting grooves, ribs, or both on a PCB are used to help increase these specifications.
- (2) This coupler is suitable for *safe electrical insulation* only within the safety ratings. Compliance with the safety ratings shall be ensured by means of suitable protective circuits.
- (3) Testing is carried out in air to determine the surge immunity of the package.
- (4) Testing is carried in oil to determine the intrinsic surge immunity of the isolation barrier.
- (5) Apparent charge is electrical discharge caused by a partial discharge (pd).
- (6) All pins on each side of the barrier are tied together, creating a two-pin device.

7.7 Safety-Related Certifications

VDE	UL
DIN EN IEC 60747-17 (VDE 0884-17), EN IEC 60747-17, DIN EN IEC 62368-1 (VDE 0868-1), EN IEC 62368-1, IEC 62368-1 Clause : 5.4.3 ; 5.4.4.4 ; 5.4.9	Recognized under 1577 component recognition
Reinforced insulation	Single protection
Certificate number: 40040142	File number: E181974

7.8 Safety Limiting Values

Safety limiting⁽¹⁾ intends to minimize potential damage to the isolation barrier upon failure of input or output circuitry. A failure of the I/O can allow low resistance to ground or the supply and, without current limiting, dissipate sufficient power to over-heat the die and damage the isolation barrier potentially leading to secondary system failures.

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
I _S	Safety input, output, or supply current	R _{θJA} = 84.6°C/W, VDDx = 5.5 V, T _J = 150°C, T _A = 25°C			268	mA
		R _{θJA} = 84.6°C/W, VDDx = 3.6 V, T _J = 150°C, T _A = 25°C, AMC1311B only			410	
P _S	Safety input, output, or total power	R _{θJA} = 84.6°C/W, T _J = 150°C, T _A = 25°C			1477	mW
T _S	Maximum safety temperature				150	°C

- (1) The maximum safety temperature, T_S, has the same value as the maximum junction temperature, T_J, specified for the device. The I_S and P_S parameters represent the safety current and safety power, respectively. Do not exceed the maximum limits of I_S and P_S. These limits vary with the ambient temperature, T_A.

The junction-to-air thermal resistance, R_{θJA}, in the [Thermal Information](#) table is that of a device installed on a high-K test board for leaded surface-mount packages. Use these equations to calculate the value for each parameter:

T_J = T_A + R_{θJA} × P, where P is the power dissipated in the device.

T_{J(max)} = T_S = T_A + R_{θJA} × P_S, where T_{J(max)} is the maximum junction temperature.

P_S = I_S × VDD_{max}, where VDD_{max} is the maximum supply voltage for high-side and low-side.

7.9 Electrical Characteristics

minimum and maximum specifications of the AMC1311 apply from $T_A = -40^{\circ}\text{C}$ to $+125^{\circ}\text{C}$, $V_{DD1} = 4.5\text{ V}$ to 5.5 V , $V_{DD2} = 3.0\text{ V}$ to 5.5 V , $V_{IN} = -0.1\text{ V}$ to 2 V , and $\text{SHTDN} = \text{GND1} = 0\text{ V}$; minimum and maximum specifications of the AMC1311B apply from $T_A = -55^{\circ}\text{C}$ to $+125^{\circ}\text{C}$, $V_{DD1} = 3.0\text{ V}$ to 5.5 V , $V_{DD2} = 3.0\text{ V}$ to 5.5 V , $V_{IN} = -0.1\text{ V}$ to 2 V , and $\text{SHTDN} = \text{GND1} = 0\text{ V}$ (unless otherwise noted); typical specifications are at $T_A = 25^{\circ}\text{C}$, $V_{DD1} = 5\text{ V}$, and $V_{DD2} = 3.3\text{ V}$

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
ANALOG INPUT						
V_{OS}	Input offset voltage ^{(1) (2)}	$T_A = 25^{\circ}\text{C}$, $4.5\text{ V} \leq V_{DD1} \leq 5.5\text{ V}$, AMC1311	-9.9	± 0.4	9.9	mV
		$T_A = 25^{\circ}\text{C}$, AMC1311B ⁽³⁾	-1.5	± 0.4	1.5	
TCV_{OS}	Input offset thermal drift ^{(1) (2) (5)}	AMC1311		± 20		$\mu\text{V}/^{\circ}\text{C}$
		AMC1311B	-10	± 3	10	
R_{IN}	Input resistance	$T_A = 25^{\circ}\text{C}$		1		G Ω
I_{IB}	Input bias current	$I_N = \text{GND1}$, $T_A = 25^{\circ}\text{C}$	-15	3.5	15	nA
C_{IN}	Input capacitance	$f_{IN} = 275\text{ kHz}$		7		pF
ANALOG OUTPUT						
	Nominal gain			1		V/V
E_G	Gain error ⁽¹⁾	$T_A = 25^{\circ}\text{C}$, AMC1311	-1%	0.4%	1%	
		$T_A = 25^{\circ}\text{C}$, AMC1311B	-0.2%	$\pm 0.05\%$	0.2%	
TCE_G	Gain error drift ^{(1) (6)}	AMC1311		± 30		ppm/ $^{\circ}\text{C}$
		AMC1311B	-40	± 5	40	
	Nonlinearity ⁽¹⁾		-0.04%	$\pm 0.01\%$	0.04%	
THD	Total harmonic distortion ⁽⁴⁾	$V_{IN} = 2 V_{PP}$, $V_{IN} > 0\text{ V}$, $f_{IN} = 10\text{ kHz}$, BW = 10 kHz		-87		dB
SNR	Signal-to-noise ratio	$V_{IN} = 2 V_{PP}$, $f_{IN} = 1\text{ kHz}$, BW = 10 kHz	79	82.6		dB
		$V_{IN} = 2 V_{PP}$, $f_{IN} = 10\text{ kHz}$, BW = 100 kHz		70.9		
	Output noise	$V_{IN} = \text{GND1}$, BW = 100 kHz		220		μV_{rms}
PSRR	Power-supply rejection ratio ⁽²⁾	vs V_{DD1} , at DC		-80		dB
		vs V_{DD2} , at DC		-85		
		vs V_{DD1} , 10 kHz / 100-mV ripple		-65		
		vs V_{DD2} , 10 kHz / 100-mV ripple		-70		
V_{CMout}	Output common-mode voltage		1.39	1.44	1.49	V
$V_{CLIPout}$	Clipping differential output voltage	$V_{OUT} = (V_{OUTP} - V_{OUTN})$; $V_{IN} > V_{Clipping}$		2.49		V
$V_{FAILSAFE}$	Failsafe differential output voltage	SHTDN = high, or V_{DD1} undervoltage, or V_{DD1} missing		-2.6	-2.5	V
BW	Output bandwidth	AMC1311	100	220		kHz
		AMC1311B	220	275		
R_{OUT}	Output resistance	On OUTP or OUTN		<0.2		Ω
	Output short-circuit current	On OUTP or OUTN, sourcing or sinking, $I_N = \text{GND1}$, outputs shorted to either GND or V_{DD2}		14		mA
CMTI	Common-mode transient immunity	AMC1311	15	30		kV/ μs
		AMC1311B	100	150		

7.9 Electrical Characteristics (continued)

minimum and maximum specifications of the AMC1311 apply from $T_A = -40^{\circ}\text{C}$ to $+125^{\circ}\text{C}$, $V_{DD1} = 4.5\text{ V}$ to 5.5 V , $V_{DD2} = 3.0\text{ V}$ to 5.5 V , $V_{IN} = -0.1\text{ V}$ to 2 V , and $\text{SHTDN} = \text{GND1} = 0\text{ V}$; minimum and maximum specifications of the AMC1311B apply from $T_A = -55^{\circ}\text{C}$ to $+125^{\circ}\text{C}$, $V_{DD1} = 3.0\text{ V}$ to 5.5 V , $V_{DD2} = 3.0\text{ V}$ to 5.5 V , $V_{IN} = -0.1\text{ V}$ to 2 V , and $\text{SHTDN} = \text{GND1} = 0\text{ V}$ (unless otherwise noted); typical specifications are at $T_A = 25^{\circ}\text{C}$, $V_{DD1} = 5\text{ V}$, and $V_{DD2} = 3.3\text{ V}$

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
DIGITAL INPUT						
I_{IN}	Input current	$\text{SHTDN pin}, \text{GND1} \leq \text{SHTDN} \leq V_{DD1}$	-70		1	μA
C_{IN}	Input capacitance	SHTDN pin		5		pF
V_{IH}	High-level input voltage		$0.7 \times V_{DD1}$			V
V_{IL}	Low-level input voltage				$0.3 \times V_{DD1}$	V
POWER SUPPLY						
$V_{DD1_{UV}}$	VDD1 undervoltage detection threshold	VDD1 rising	2.5	2.7	2.9	V
		VDD1 falling	2.4	2.6	2.8	
$V_{DD2_{UV}}$	VDD2 undervoltage detection threshold	VDD2 rising	2.2	2.45	2.65	V
		VDD2 falling	1.85	2.0	2.2	
I_{DD1}	High-side supply current	$3.0\text{ V} < V_{DD1} < 3.6\text{ V}$, $\text{SHTDN} = \text{low}$, AMC1311B only		6.0	8.4	mA
		$4.5\text{ V} < V_{DD1} < 5.5\text{ V}$, $\text{SHTDN} = \text{low}$		7.1	9.7	
		$\text{SHTDN} = V_{DD1}$		1.3		μA
I_{DD2}	Low-side supply current	$3.0\text{ V} < V_{DD2} < 3.6\text{ V}$		5.3	7.2	mA
		$4.5\text{ V} < V_{DD2} < 5.5\text{ V}$		5.9	8.1	

- (1) The typical value includes one standard deviation (*sigma*) at nominal operating conditions.
- (2) This parameter is input referred.
- (3) The typical value is at $V_{DD1} = 3.3\text{ V}$.
- (4) THD is the ratio of the rms sum of the amplitudes of first five higher harmonics to the amplitude of the fundamental.
- (5) Offset error temperature drift is calculated using the box method, as described by the following equation:

$$TCV_{OS} = (Value_{MAX} - Value_{MIN}) / TempRange$$
- (6) Gain error temperature drift is calculated using the box method, as described by the following equation:

$$TCE_G (\text{ppm}) = (Value_{MAX} - Value_{MIN}) / (Value_{(T=25^{\circ}\text{C})} \times TempRange) \times 10^6$$

7.10 Switching Characteristics

over operating ambient temperature range (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
t_r	Output signal rise time			1.3		μs
t_f	Output signal fall time			1.3		μs
	IN to OUTx signal delay (50% – 10%)	Unfiltered output, AMC1311		1.5	2.5	μs
		Unfiltered output, AMC1311B		1.0	1.5	
	IN to OUTx signal delay (50% – 50%)	Unfiltered output, AMC1311		2.1	3.1	μs
		Unfiltered output, AMC1311B		1.6	2.1	
	IN to OUTx signal delay (50% – 90%)	Unfiltered output, AMC1311		3.0	4.0	μs
		Unfiltered output, AMC1311B		2.5	3.0	
t_{AS}	Analog settling time	VDD1 step to 3.0 V with VDD2 $\geq$ 3.0 V, to V_{OUTP} , V_{OUTN} valid, 0.1% settling		50	100	μs
t_{EN}	Device enable time	SHTDN high to low		50	100	μs
t_{SHTDN}	Device shutdown time	SHTDN low to high		3	10	μs

7.11 Timing Diagram

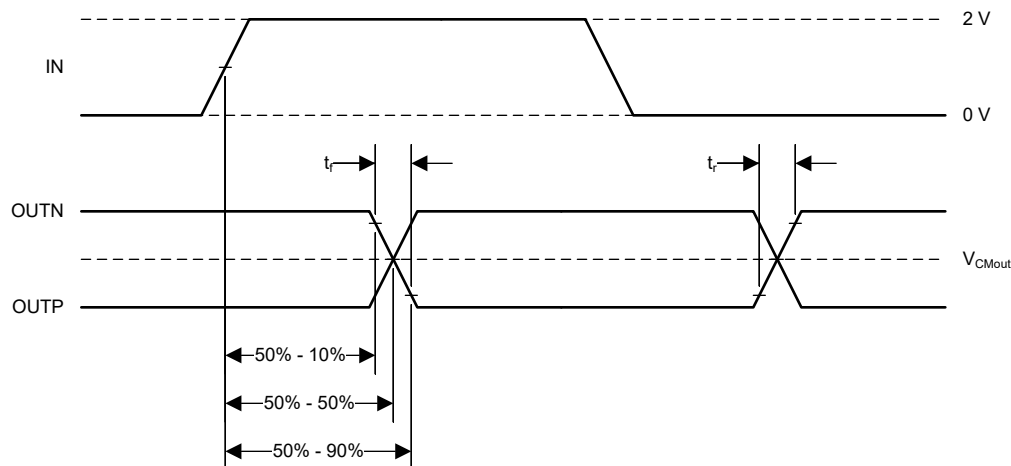


Figure 7-1. Rise, Fall, and Delay Time Definition

7.12 Insulation Characteristics Curves

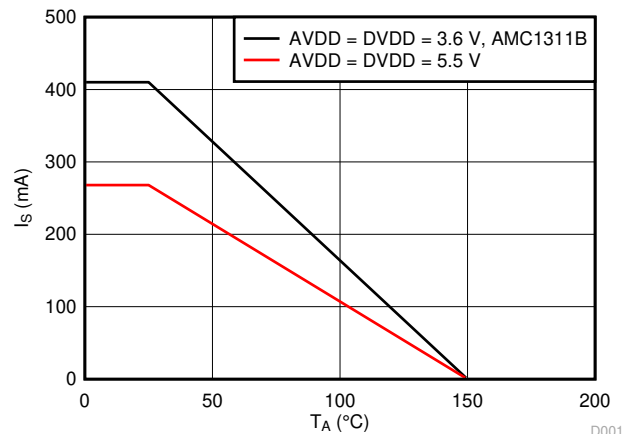


Figure 7-2. Thermal Derating Curve for Safety-Limiting Current per VDE

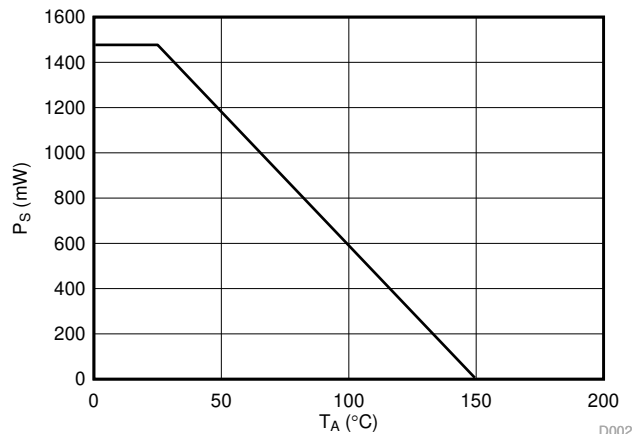
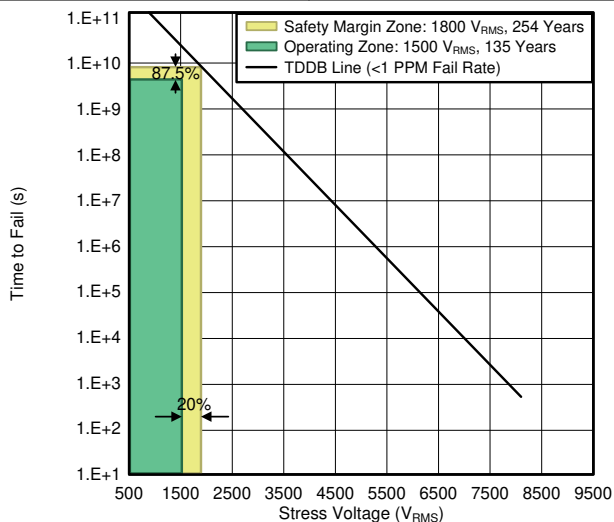


Figure 7-3. Thermal Derating Curve for Safety-Limiting Power per VDE

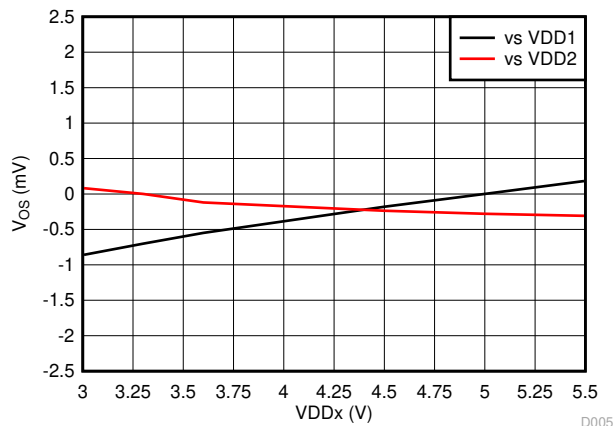


T_A up to 150°C, stress-voltage frequency = 60 Hz, isolation working voltage = 1500 V_{RMS} , operating lifetime = 135 years

Figure 7-4. Reinforced Isolation Capacitor Lifetime Projection

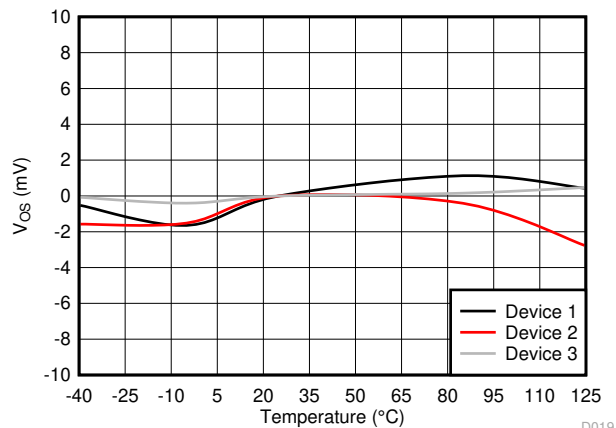
7.13 Typical Characteristics

at VDD1 = 5 V, VDD2 = 3.3 V, SHTDN = 0 V, f_{IN} = 10 kHz, and BW = 100 kHz (unless otherwise noted)



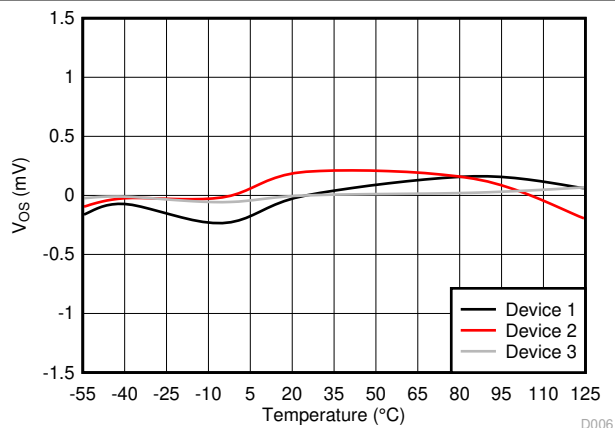
3.0 V ≤ VDD1 < 4.5 V for the AMC1311B only

Figure 7-5. Input Offset Voltage vs Supply Voltage



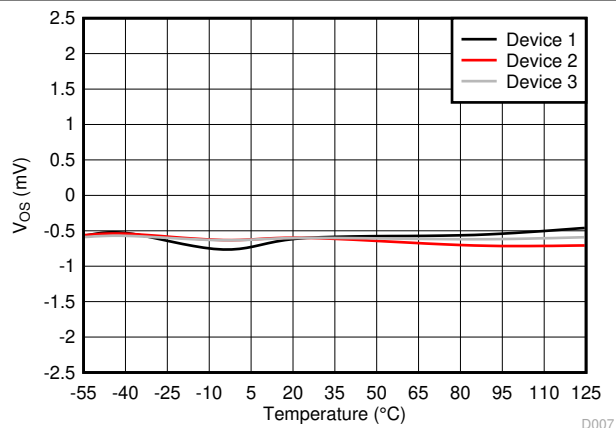
AMC1311

Figure 7-6. Input Offset Voltage vs Temperature



VDD1 = 5 V, AMC1311B

Figure 7-7. Input Offset Voltage vs Temperature



VDD1 = 3.3 V, AMC1311B

Figure 7-8. Input Offset Voltage vs Temperature

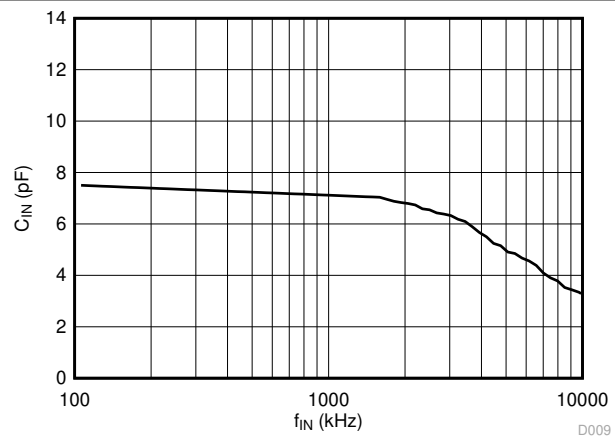
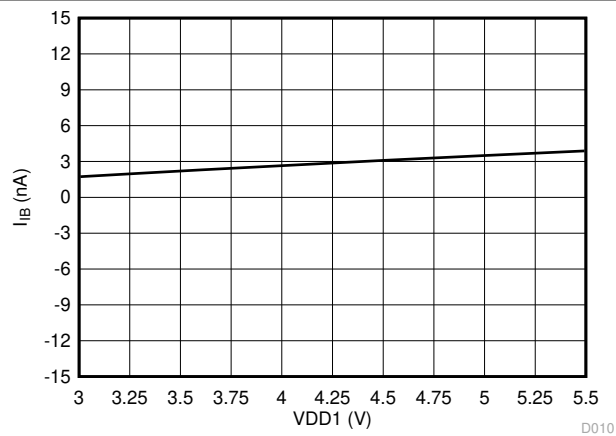


Figure 7-9. Input Capacitance vs Input Signal Frequency



3.0 V ≤ VDD1 < 4.5 V for the AMC1311B only

Figure 7-10. Input Bias Current vs High-Side Supply Voltage

7.13 Typical Characteristics (continued)

at VDD1 = 5 V, VDD2 = 3.3 V, SHTDN = 0 V, f_{IN} = 10 kHz, and BW = 100 kHz (unless otherwise noted)

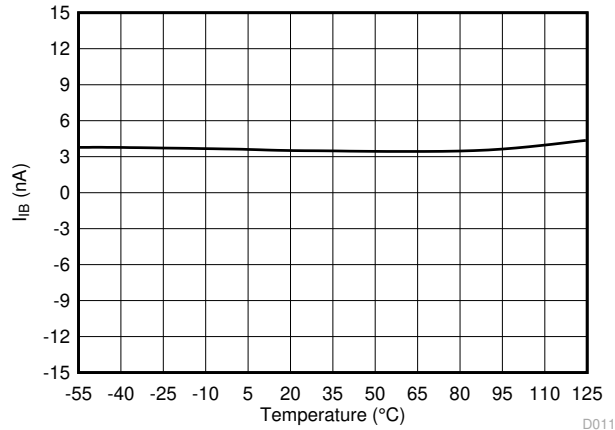


Figure 7-11. Input Bias Current vs Temperature

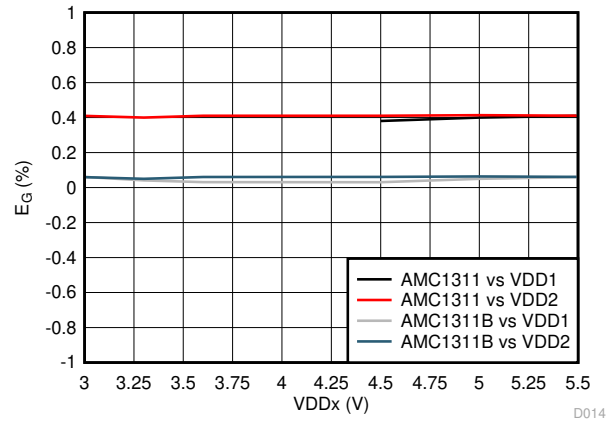


Figure 7-12. Gain Error vs Supply Voltage

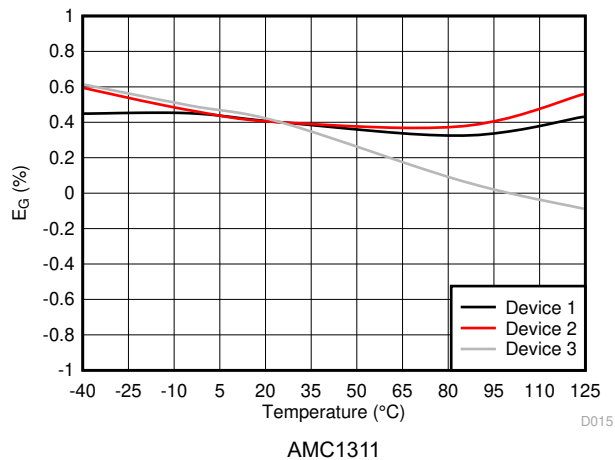


Figure 7-13. Gain Error vs Temperature

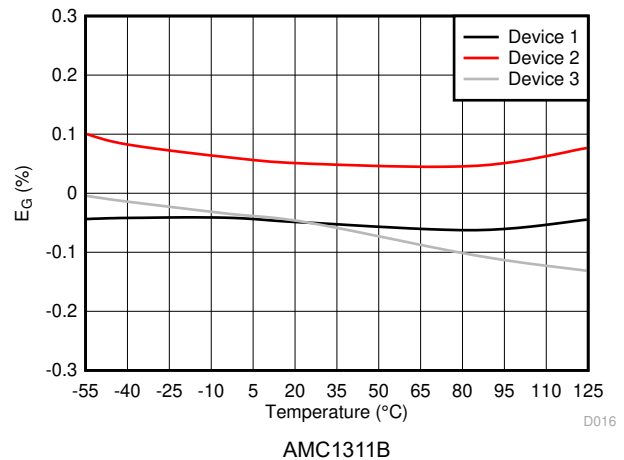


Figure 7-14. Gain Error vs Temperature

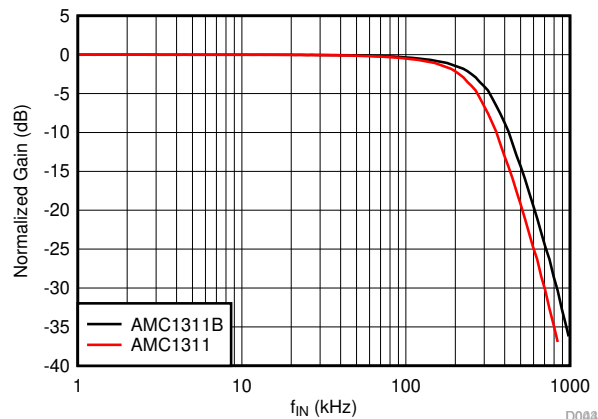


Figure 7-15. Normalized Gain vs Input Frequency

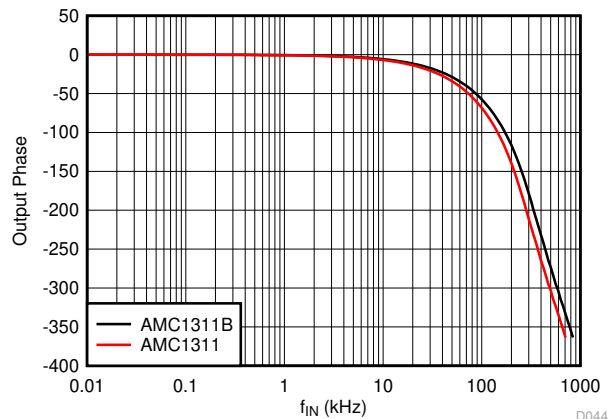


Figure 7-16. Output Phase vs Input Frequency

7.13 Typical Characteristics (continued)

at VDD1 = 5 V, VDD2 = 3.3 V, SHTDN = 0 V, f_{IN} = 10 kHz, and BW = 100 kHz (unless otherwise noted)

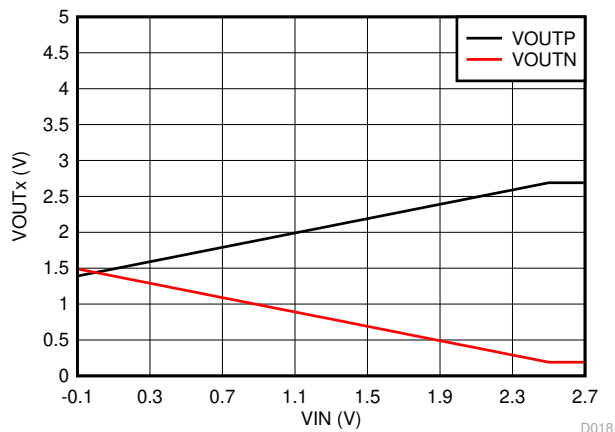


Figure 7-17. Output Voltage vs Input Voltage

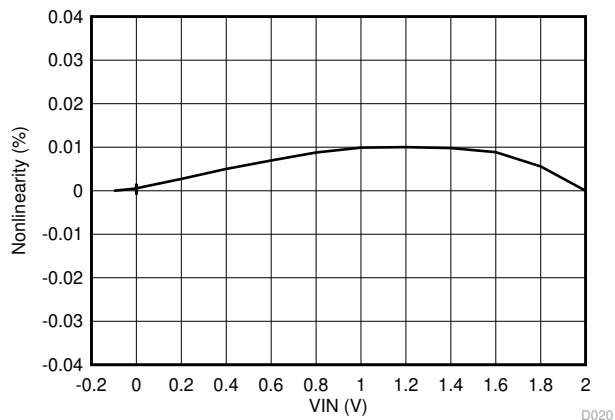
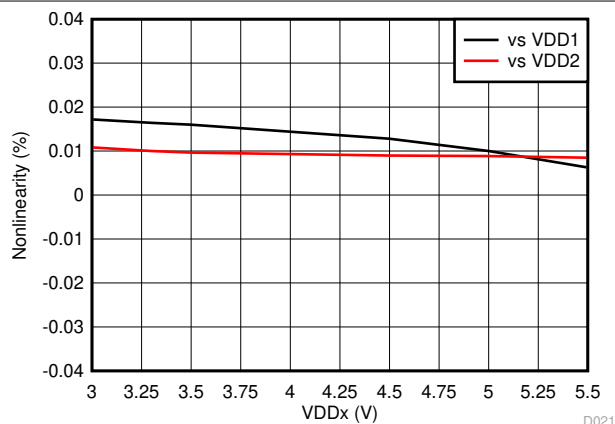
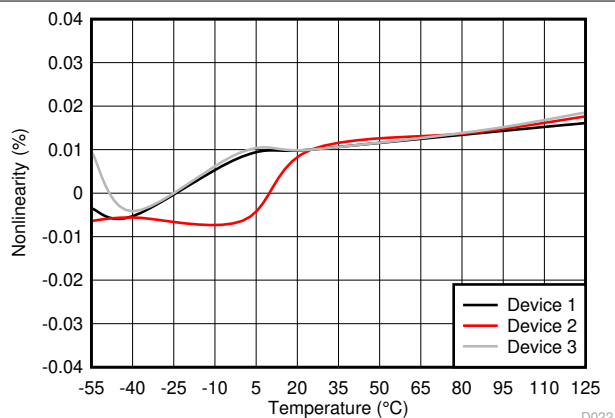


Figure 7-18. Nonlinearity vs Input Voltage



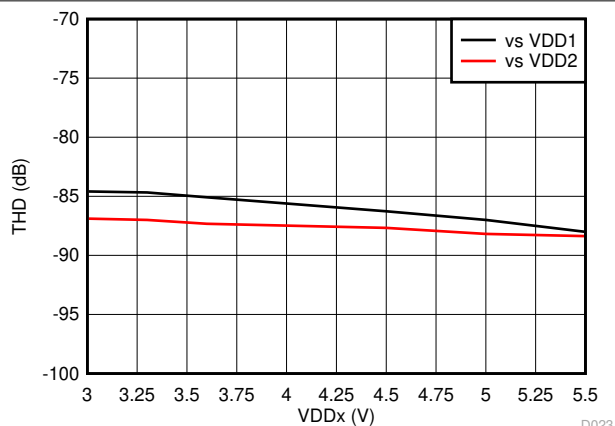
3.0 V ≤ VDD1 < 4.5 V for the AMC1311B only

Figure 7-19. Nonlinearity vs Supply Voltage



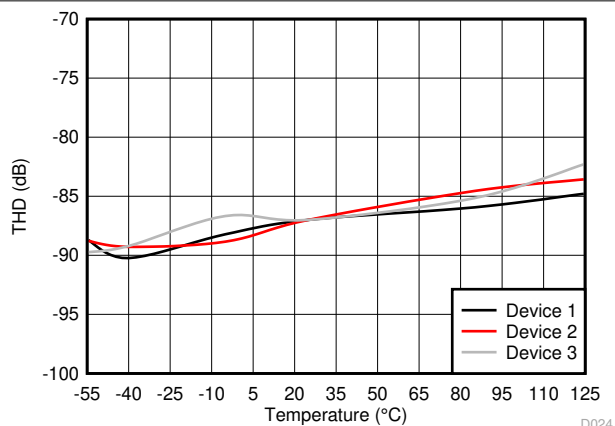
-55°C ≤ T_A < -40°C for the AMC1311B only

Figure 7-20. Nonlinearity vs Temperature



3.0 V ≤ VDD1 < 4.5 V for the AMC1311B only

Figure 7-21. Total Harmonic Distortion vs Supply Voltage



-55°C ≤ T_A < -40°C for the AMC1311B only

Figure 7-22. Total Harmonic Distortion vs Temperature

7.13 Typical Characteristics (continued)

at VDD1 = 5 V, VDD2 = 3.3 V, SHTDN = 0 V, f_{IN} = 10 kHz, and BW = 100 kHz (unless otherwise noted)

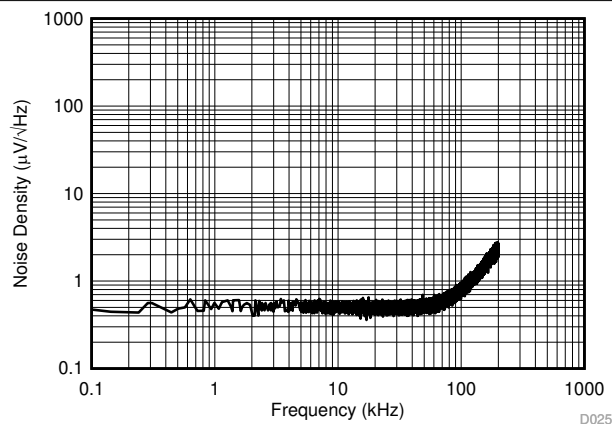


Figure 7-23. Input-Referred Noise Density vs Frequency

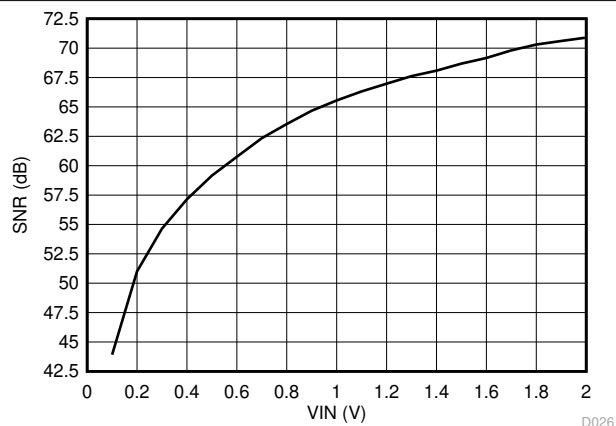
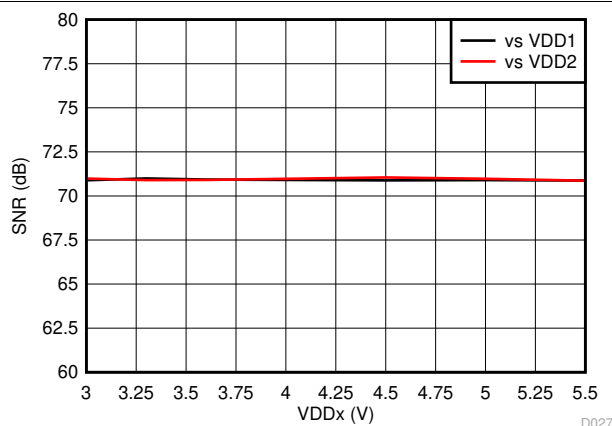
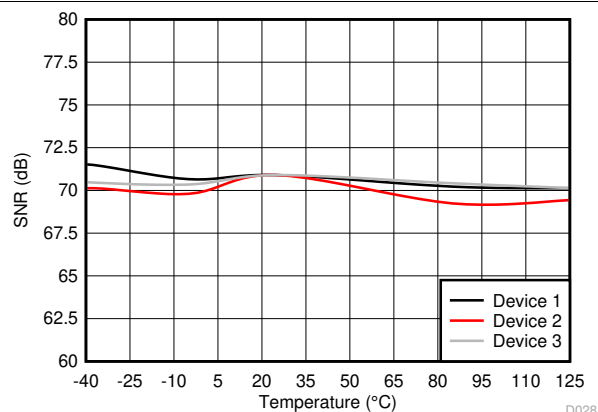


Figure 7-24. Signal-to-Noise Ratio vs Input Voltage



3.0 V ≤ VDD1 < 4.5 V for the AMC1311B only

Figure 7-25. Signal-to-Noise Ratio vs Supply Voltage



–55°C ≤ T_A < –40°C for the AMC1311B only

Figure 7-26. Signal-to-Noise Ratio vs Temperature

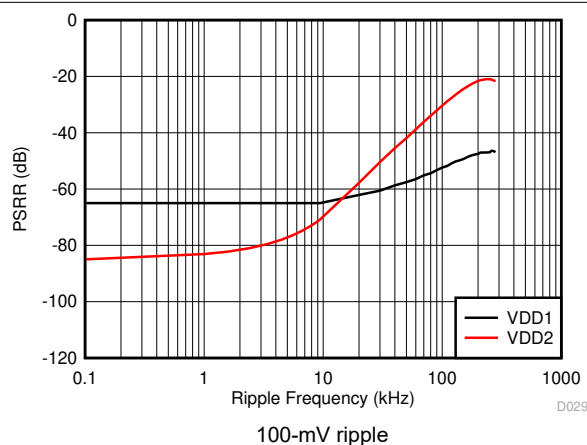


Figure 7-27. Power-Supply Rejection Ratio vs Ripple Frequency

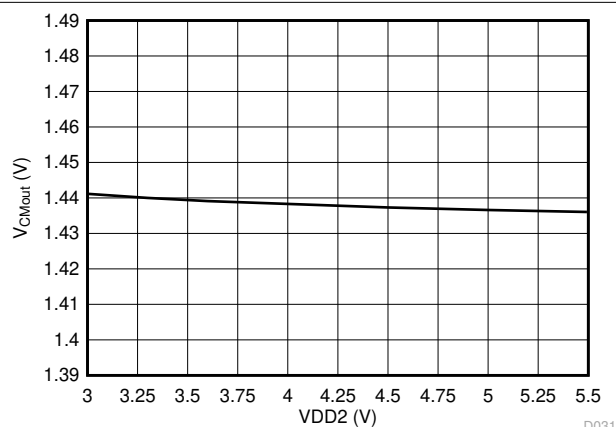


Figure 7-28. Output Common-Mode Voltage vs Low-Side Supply Voltage

7.13 Typical Characteristics (continued)

at VDD1 = 5 V, VDD2 = 3.3 V, SHTDN = 0 V, f_{IN} = 10 kHz, and BW = 100 kHz (unless otherwise noted)

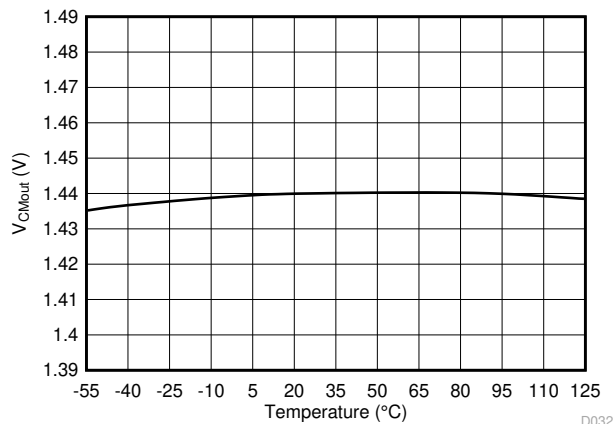


Figure 7-29. Output Common-Mode Voltage vs Temperature

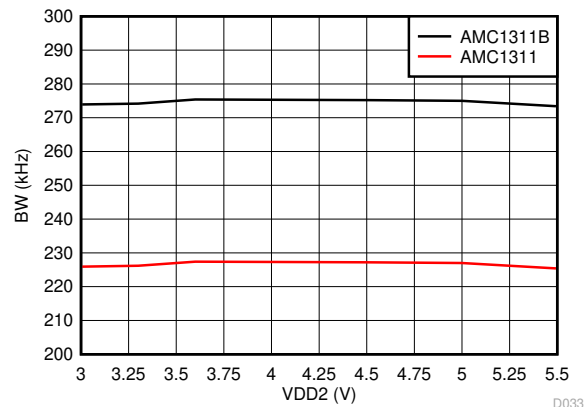


Figure 7-30. Output Bandwidth vs Low-Side Supply Voltage

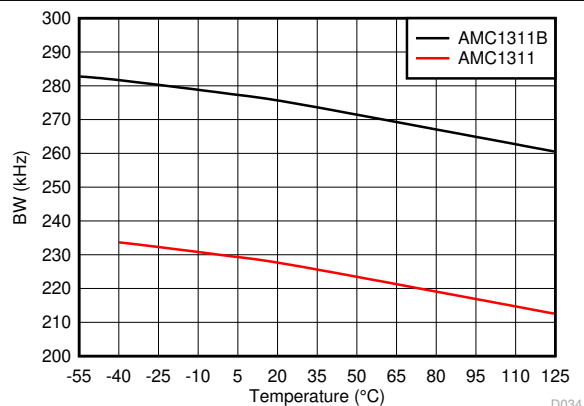


Figure 7-31. Output Bandwidth vs Temperature

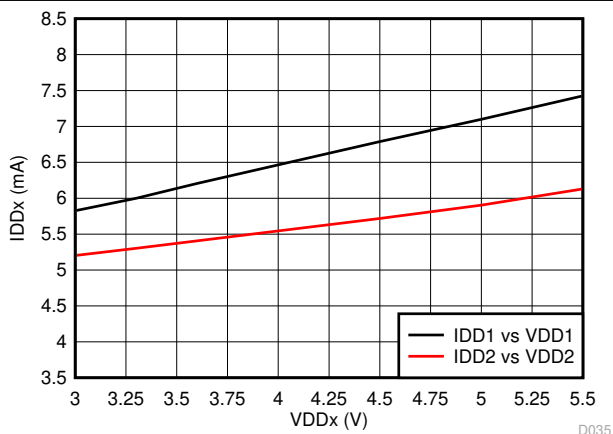


Figure 7-32. Supply Current vs Supply Voltage
3.0 V ≤ VDD1 < 4.5 V for the AMC1311B only

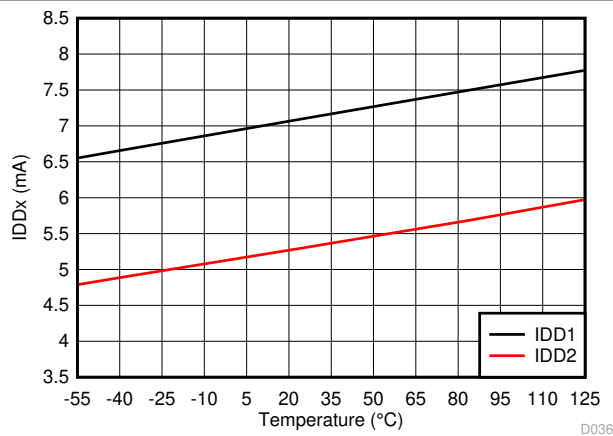


Figure 7-33. Supply Current vs Temperature
-55°C ≤ T_A < -40°C for the AMC1311B only

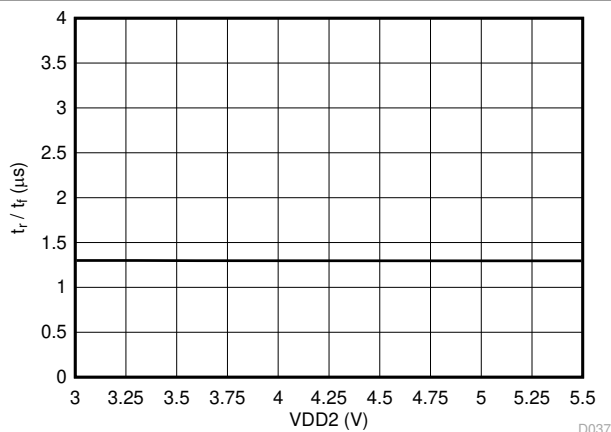
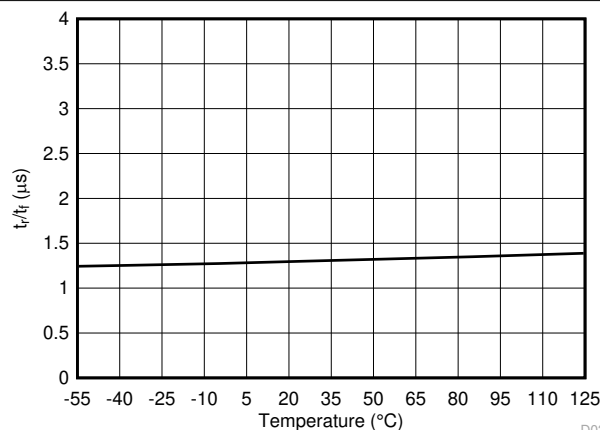


Figure 7-34. Output Rise and Fall Time vs Low-Side Supply Voltage

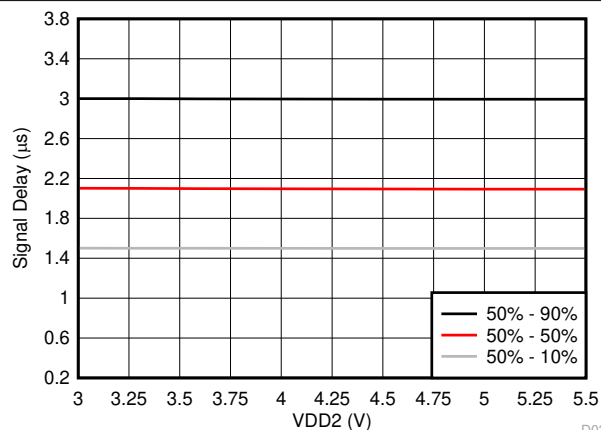
7.13 Typical Characteristics (continued)

at VDD1 = 5 V, VDD2 = 3.3 V, SHTDN = 0 V, f_{IN} = 10 kHz, and BW = 100 kHz (unless otherwise noted)



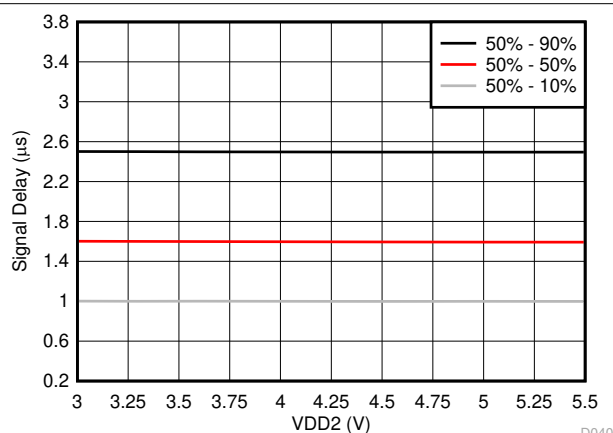
–55°C $\leq T_A < -40^\circ\text{C}$ for the AMC1311B only

Figure 7-35. Output Rise and Fall Time vs Temperature



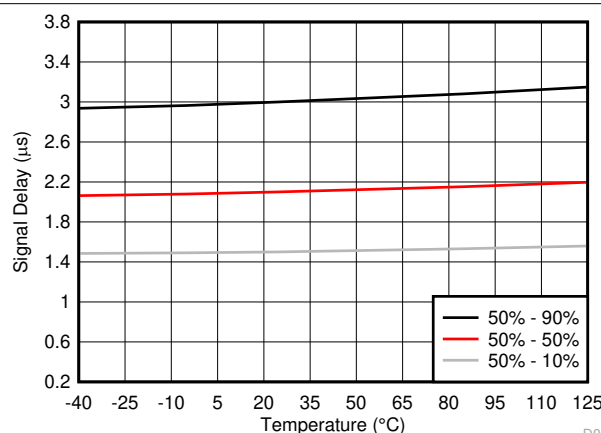
AMC1311

Figure 7-36. IN to OUTP, OUTN Signal Delay vs Low-Side Supply Voltage



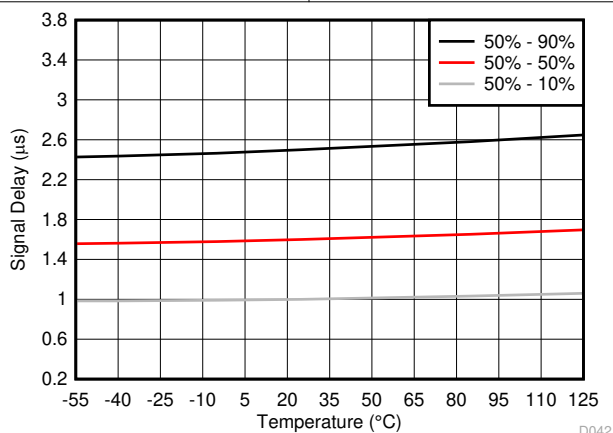
AMC1311B

Figure 7-37. IN to OUTP, OUTN Signal Delay vs Low-Side Supply Voltage



AMC1311

Figure 7-38. IN to OUTP, OUTN Signal Delay vs Temperature



AMC1311B

Figure 7-39. IN to OUTP, OUTN Signal Delay vs Temperature

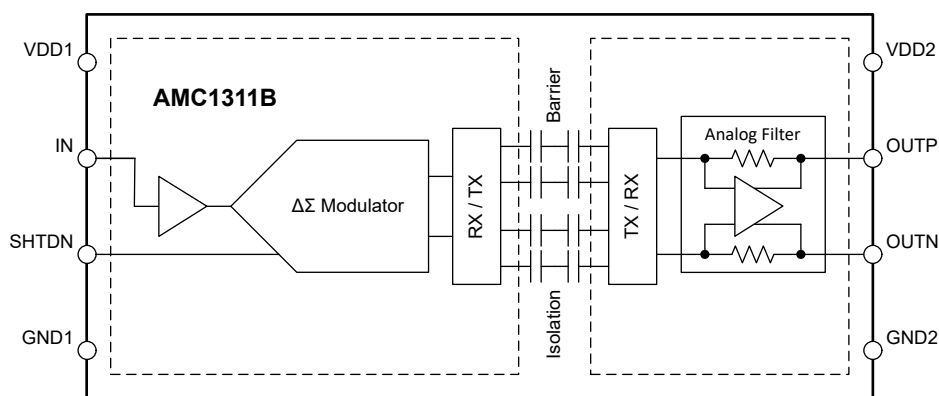
8 Detailed Description

8.1 Overview

The AMC1311 is a precision, single-ended input, isolated amplifier with a high input impedance and wide input voltage range. The input stage of the device drives a second-order, delta-sigma ($\Delta\Sigma$) modulator. The modulator converts the analog input signal into a digital bitstream that is transferred across the isolation barrier and separates the high-side from the low-side. On the low-side, the received bitstream is processed by a fourth-order analog filter that outputs a differential signal at the OUTP and OUTN pins proportional to the input signal.

The SiO₂-based, capacitive isolation barrier supports a high level of magnetic field immunity, as described in the [ISO72x Digital Isolator Magnetic-Field Immunity application report](#). The digital modulation used in the AMC1311 to transmit data across the isolation barrier, and the isolation barrier characteristics itself, result in high reliability and high common-mode transient immunity.

8.2 Functional Block Diagram



8.3 Feature Description

8.3.1 Analog Input

The single-ended, high-impedance input stage of the AMC1311 feeds a second-order, switched-capacitor, feed-forward $\Delta\Sigma$ modulator. The modulator converts the analog signal into a bitstream that is transferred across the isolation barrier, as described in the [Isolation Channel Signal Transmission](#) section.

There are two restrictions on the analog input signal IN. First, if the input voltage V_{IN} exceeds the range specified in the [Absolute Maximum Ratings](#) table, the input current must be limited to the absolute maximum value because the electrostatic discharge (ESD) protection turns on. Secondly, the linearity and parametric performance of the device is ensured only when the analog input voltage remains within the linear full-scale range (V_{FSR}) as specified in the [Recommended Operating Conditions](#) table.

8.3.2 Isolation Channel Signal Transmission

The AMC1311 uses an on-off keying (OOK) modulation scheme, as shown in [Figure 8-1](#), to transmit the modulator output bitstream across the SiO₂-based isolation barrier. The transmit driver (TX) shown in the [Functional Block Diagram](#) transmits an internally-generated, high-frequency carrier across the isolation barrier to represent a digital *one* and does not send a signal to represent a digital *zero*. The nominal frequency of the carrier used inside the AMC1311 is 480 MHz.

The receiver (RX) on the other side of the isolation barrier recovers and demodulates the signal and provides the input to the fourth-order analog filter. The AMC1311 transmission channel is optimized to achieve the highest level of common-mode transient immunity (CMTI) and lowest level of radiated emissions caused by the high-frequency carrier and RX/TX buffer switching.

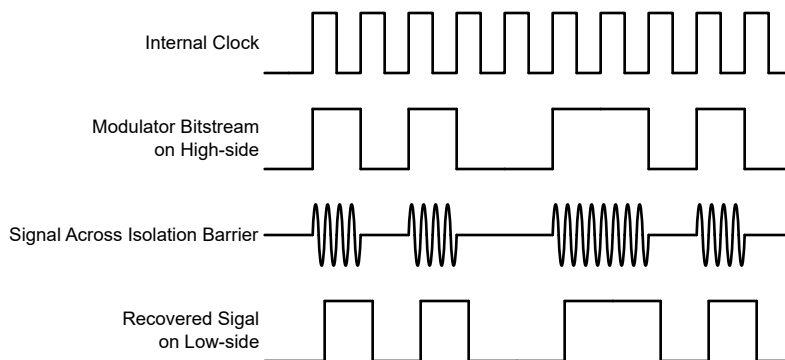


Figure 8-1. OOK-Based Modulation Scheme

8.3.3 Analog Output

The AMC1311 provides a differential analog output on the OUTP and OUTN pins. For input voltages of V_{IN} in the range from -0.1 V to $+2\text{ V}$, the device provides a linear response with a nominal gain of 1. For example, for an input voltage of 2 V , the differential output voltage ($V_{OUTP} - V_{OUTN}$) is 2 V . At zero input (IN shorted to GND1), both pins output the same common-mode output voltage V_{CMout} , as specified in the [Electrical Characteristics](#) table. For input voltages greater than 2 V but less than approximately 2.5 V , the differential output voltage continues to increase but with reduced linearity performance. The outputs saturate at a differential output voltage of $V_{CLIPout}$, as shown in [Figure 8-2](#), if the input voltage exceeds the $V_{Clipping}$ value.

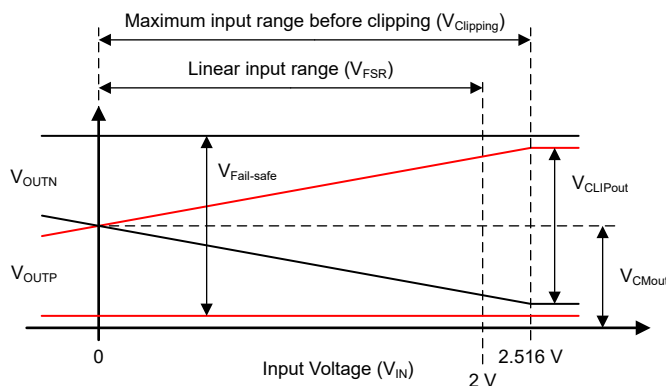


Figure 8-2. Output Behavior of the AMC1311

The AMC1311 output offers a fail-safe feature that simplifies diagnostics on a system level. [Figure 8-2](#) shows the fail-safe mode, in which the AMC1311 outputs a negative differential output voltage that does not occur under normal operating conditions. The fail-safe output is active in three cases:

- When the high-side supply VDD1 of the AMC1311 device is missing
- When the high-side supply VDD1 falls below the undervoltage threshold $VDD1_{UV}$
- When the SHTDN pin is pulled high

Use the maximum $V_{Fail-safe}$ voltage specified in the [Electrical Characteristics](#) table as a reference value for fail-safe detection on a system level.

8.4 Device Functional Modes

The AMC1311 is operational when the power supplies VDD1 and VDD2 are applied, as specified in the [Recommended Operating Conditions](#) table.

9 Application and Implementation

Note

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes, as well as validating and testing their design implementation to confirm system functionality.

9.1 Application Information

The high input impedance, low input bias current, low AC and DC errors, and low temperature drift make the AMC1311 a high-performance solution for industrial applications where voltage sensing in the presence of high common-mode voltage levels is required.

9.2 Typical Application

Figure 9-1 shows the AMC1311 in a typical application. The DC bus voltage is divided down to an approximate 2-V level across the bottom resistor (RSNS) of a high-impedance resistive divider that is sensed by the AMC1311. The AMC1311 digitizes the analog input signal on the high-side, transfers the data across the isolation barrier to the low-side, reconstructs the analog signal, and presents this signal as a differential voltage signal on the output pins.

The high-impedance input and the high common-mode transient immunity (CMTI) of the AMC1311 ensure reliable and accurate operation even in high-noise environments.

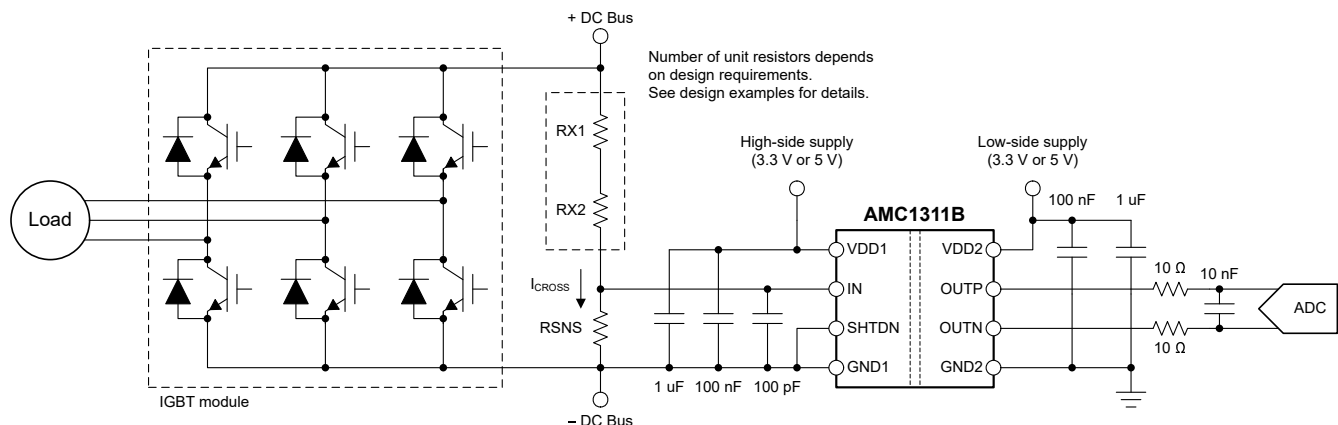


Figure 9-1. Using the AMC1311 for DC Link Voltage Sensing in Frequency Inverters

9.2.1 Design Requirements

Table 9-1 lists the parameters for this typical application.

Table 9-1. Design Requirements

PARAMETER	VALUE
System input voltage	Single phase, 230 V, 50 Hz
Maximum DC link voltage	400 V
High-side supply voltage	3.3 V or 5 V
Low-side supply voltage	3.3 V or 5 V
Maximum resistor operating voltage	75 V
Voltage drop across the sense resistor (RSNS) for a linear response	2 V (maximum)
Current through the resistive divider, I _{CROSS}	100 μ A (maximum)

9.2.2 Detailed Design Procedure

The 100- μ A, cross-current requirement at the maximum DC link voltage (400 V) determines that the total impedance of the resistive divider is 4 M Ω . The impedance of the resistive divider is dominated by the top portion (shown exemplary as RX1 and RX2 in Figure 9-1) and the voltage drop across RSNS can be neglected for a moment. The maximum allowed voltage drop per unit resistor is specified as 75 V; therefore, the minimum number of unit resistors in the top portion of the resistive divider is $400 \text{ V} / 75 \text{ V} = 6$. The calculated unit value is $4 \text{ M}\Omega / 6 = 666 \text{ k}\Omega$ and the next closest value from the E96 series is 665 k Ω .

RSNS is sized such that the voltage drop across the resistor at the maximum DC link voltage (400 V) equals the linear full-scale range input voltage (V_{FSR}) of the AMC1311, which is 2 V. This voltage is calculated as $RSNS = V_{FSR} / (V_{DC-link, max} - V_{FSR}) \times R_{TOP}$, where R_{TOP} is the total value of the top resistor string ($6 \times 665 \text{ k}\Omega = 3990 \text{ k}\Omega$). RSNS is calculated as 20.05 k Ω and matches a value from the E96 series.

Table 9-2. Resistor Value Example

PARAMETER	VALUE
Unit resistor R_X	665 k Ω
Number of unit resistors	6
Sense resistor RSNS	20.05 k Ω
Resulting current through resistive divider I _{CROSS}	99.7 μ A
Resulting voltage drop across sense resistor	2.000 V
Power dissipated in unit resistor RX	6.6 mW
Total power dissipated in resistive divider	39.9 mW

9.2.2.1 Input Filter Design

Placing an RC filter in front of the isolated amplifier improves signal-to-noise performance of the signal path. In practice, however, the impedance of the resistor divider is high and only a small-value filter capacitor can be used to not limit the signal bandwidth to an unacceptable low value. Design the input filter such that:

- The cutoff frequency of the filter is at least one order of magnitude lower than the sampling frequency (20 MHz) of the internal $\Delta\Sigma$ modulator
- The input bias current does not generate significant voltage drop across the DC impedance of the input filter

Most voltage-sensing applications use high-impedance resistor dividers in front of the isolated amplifier to scale down the input voltage. In this case, a single capacitor (as shown in Figure 9-2) is sufficient to filter the input signal.

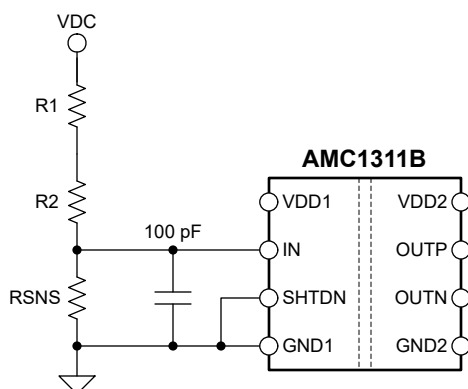


Figure 9-2. Input Filter

9.2.2.2 Differential to Single-Ended Output Conversion

Figure 9-3 shows an example of a TLV900x-based signal conversion and filter circuit for systems using single-ended input ADCs to convert the analog output voltage into digital. With $R1 = R2 = R3 = R4$, the output voltage equals $(V_{OUTP} - V_{OUTN}) + V_{REF}$. Tailor the bandwidth of this filter stage to the bandwidth requirement of the system and use NP0-type capacitors for best performance. For most applications, $R1 = R2 = R3 = R4 = 3.3 \text{ k}\Omega$ and $C1 = C2 = 330 \text{ pF}$ yields good performance.

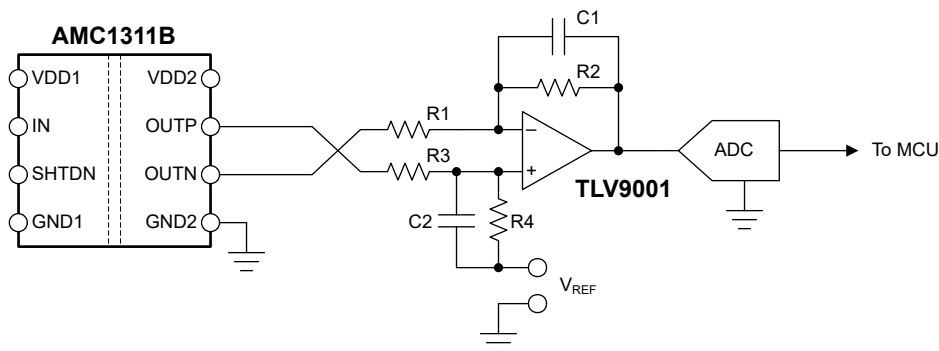


Figure 9-3. Connecting the AMC1311 Output to a Single-Ended Input ADC

For more information on the general procedure to design the filtering and driving stages of SAR ADCs, see the [18-Bit, 1MSPS Data Acquisition Block \(DAQ\) Optimized for Lowest Distortion and Noise](#) and [18-Bit Data Acquisition Block \(DAQ\) Optimized for Lowest Power reference guides](#), available for download at www.ti.com.

9.2.3 Application Curve

One important aspect of system design is the effective detection of an overvoltage condition to protect switching devices and passive components from damage. To power off the system quickly in the event of an overvoltage condition, a low delay caused by the isolated amplifier is required. Figure 9-4 shows the typical full-scale step response of the AMC1311.

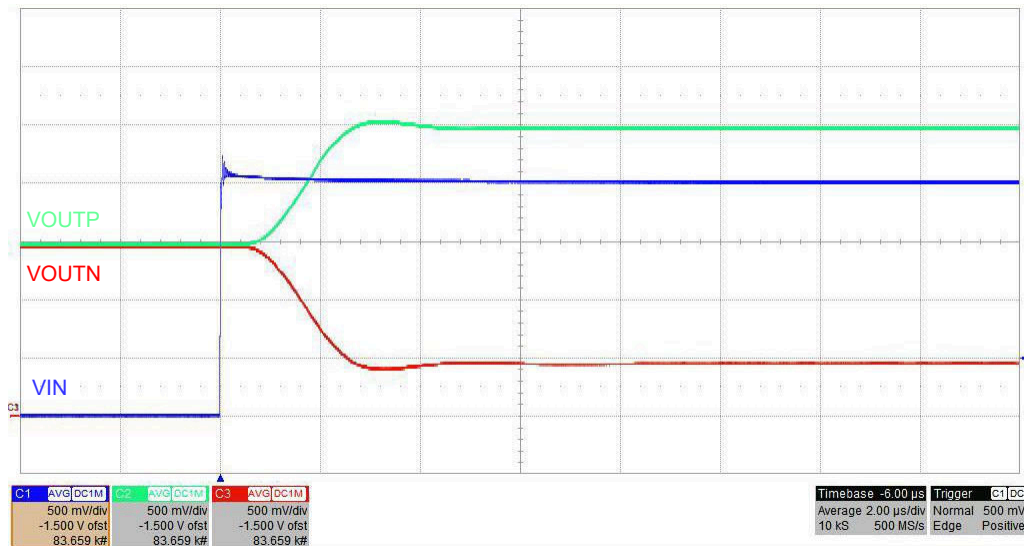


Figure 9-4. Step Response of the AMC1311

9.3 What To Do and What Not To Do

Do not leave the analog input (IN pin) of the AMC1311 unconnected (floating) when the device is powered up on the high-side. If the device input is left floating, the bias current may generate a negative input voltage that exceeds the specified input voltage range, causing the output of the device to be invalid.

Do not connect protection diodes to the input (IN pin) of the AMC1311. Diode leakage current can introduce significant measurement error especially at high temperatures. The input pin is protected against high voltages by its ESD protection circuit and the high impedance of the external resistive divider.

10 Power Supply Recommendations

In a typical application, the high-side (VDD1) of the AMC1311 is powered from an already existing, high-side, ground-referenced, 3.3-V or 5-V power supply in the system. Alternatively, the high-side supply can be generated from the low-side supply (VDD2) by an isolated DC/DC converter. A low-cost solution is based on the push-pull driver [SN6501](#) and a transformer that supports the desired isolation voltage ratings.

The AMC1311 does not require any specific power-up sequencing. The high-side power supply (VDD1) is decoupled with a low-ESR, 100-nF capacitor (C1) parallel to a low-ESR, 1-μF capacitor (C2). The low-side power supply (VDD2) is equally decoupled with a low-ESR, 100-nF capacitor (C3) parallel to a low-ESR, 1-μF capacitor (C4). Place all four capacitors (C1, C2, C3, and C4) as close to the device as possible. [Figure 10-1](#) shows the proper decoupling layout for the AMC1311.

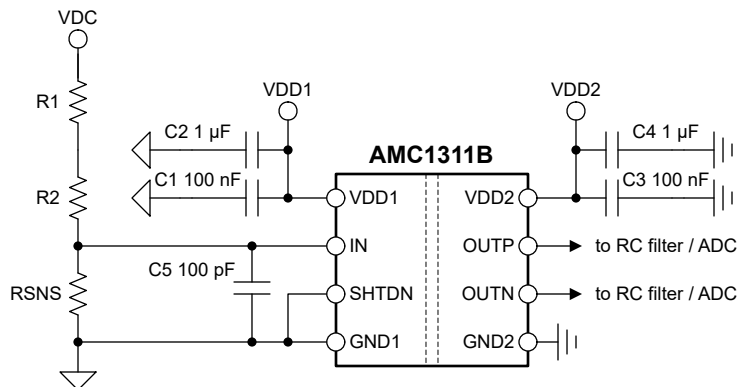


Figure 10-1. Decoupling of the AMC1311

Capacitors must provide adequate effective capacitance under the applicable DC bias conditions they experience in the application. Multilayer ceramic capacitors (MLCC) typically exhibit only a fraction of their nominal capacitance under real-world conditions and this factor must be taken into consideration when selecting these capacitors. This problem is especially acute in low-profile capacitors, in which the dielectric field strength is higher than in taller components. Reputable capacitor manufacturers provide capacitance versus DC bias curves that greatly simplify component selection.

11 Layout

11.1 Layout Guidelines

Figure 11-1 shows a layout recommendation with the critical placement of the decoupling capacitors (as close as possible to the AMC1311 supply pins) and placement of the other components required by the device. For best performance, place the sense resistor close to the device input pin (IN).

11.2 Layout Example

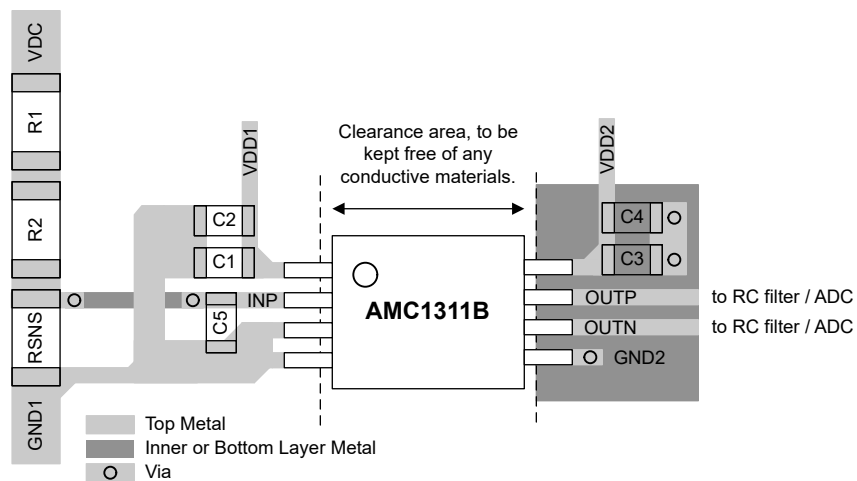


Figure 11-1. Recommended Layout of the AMC1311

12 Device and Documentation Support

12.1 Documentation Support

12.1.1 Related Documentation

For related documentation, see the following:

- Texas Instruments, [Isolation Glossary application report](#)
- Texas Instruments, [Semiconductor and IC Package Thermal Metrics application report](#)
- Texas Instruments, [ISO72x Digital Isolator Magnetic-Field Immunity application report](#)
- Texas Instruments, [TLV900x Low-Power, RRIO, 1-MHz Operational Amplifier for Cost-Sensitive Systems data sheet](#)
- Texas Instruments, [SN6501 Transformer Driver for Isolated Power Supplies data sheet](#)
- Texas Instruments, [AMC1311EVM Users Guide](#)
- Texas Instruments, [18-Bit, 1-MSPS Data Acquisition Block \(DAQ\) Optimized for Lowest Distortion and Noise reference guide](#)
- Texas Instruments, [18-Bit, 1-MSPS Data Acquisition Block \(DAQ\) Optimized for Lowest Power reference guide](#)
- Texas Instruments, [Isolated Amplifier Voltage Sensing Excel Calculator design tool](#)

12.2 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. Click on *Subscribe to updates* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

12.3 Support Resources

[TI E2E™ support forums](#) are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

Linked content is provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's [Terms of Use](#).

12.4 Trademarks

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12.5 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

12.6 Glossary

[TI Glossary](#) This glossary lists and explains terms, acronyms, and definitions.

13 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
AMC1311BDWV	Active	Production	SOIC (DWV) 8	64 TUBE	Yes	NIPDAU	Level-3-260C-168 HR	-55 to 125	1311B
AMC1311BDWVR	Active	Production	SOIC (DWV) 8	1000 LARGE T&R	Yes	NIPDAU	Level-3-260C-168 HR	-55 to 125	1311B
AMC1311DWV	Active	Production	SOIC (DWV) 8	64 TUBE	Yes	NIPDAU	Level-3-260C-168 HR	-40 to 125	1311
AMC1311DWVR	Active	Production	SOIC (DWV) 8	1000 LARGE T&R	Yes	NIPDAU	Level-3-260C-168 HR	-40 to 125	1311

⁽¹⁾ **Status:** For more details on status, see our [product life cycle](#).

⁽²⁾ **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

⁽³⁾ **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

⁽⁴⁾ **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

⁽⁵⁾ **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

⁽⁶⁾ **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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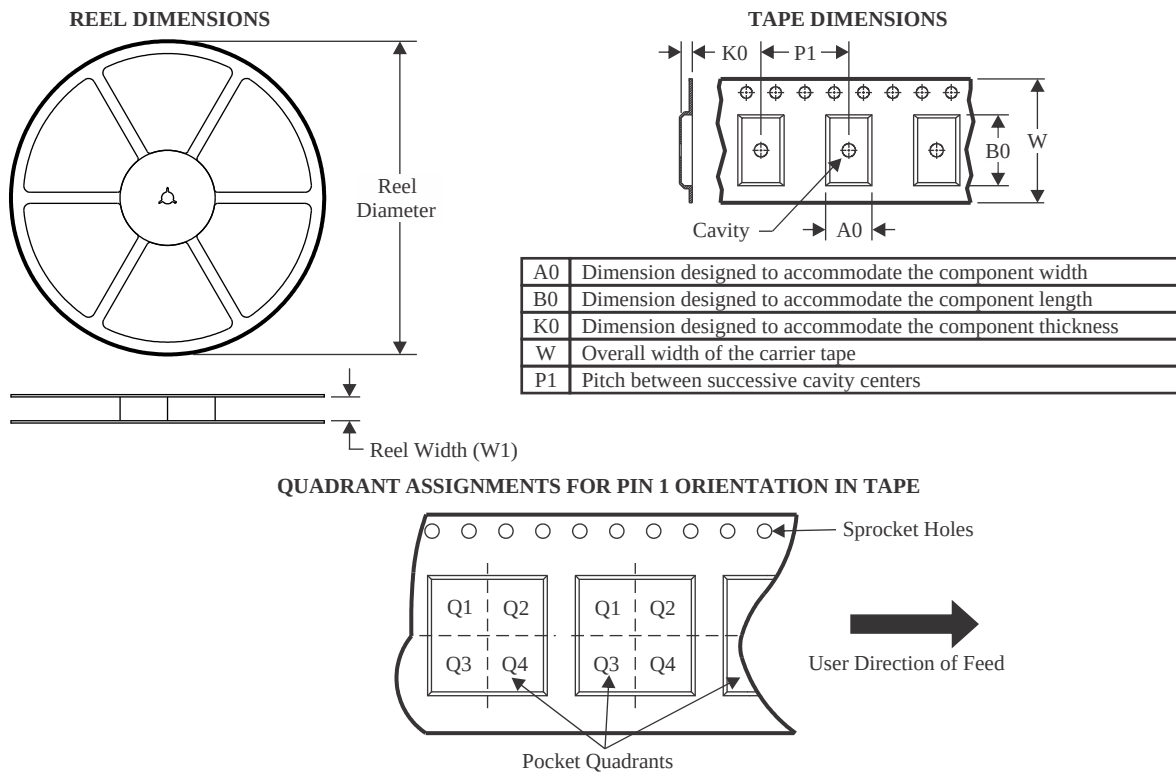
OTHER QUALIFIED VERSIONS OF AMC1311 :

- Automotive : [AMC1311-Q1](#)

NOTE: Qualified Version Definitions:

- Automotive - Q100 devices qualified for high-reliability automotive applications targeting zero defects

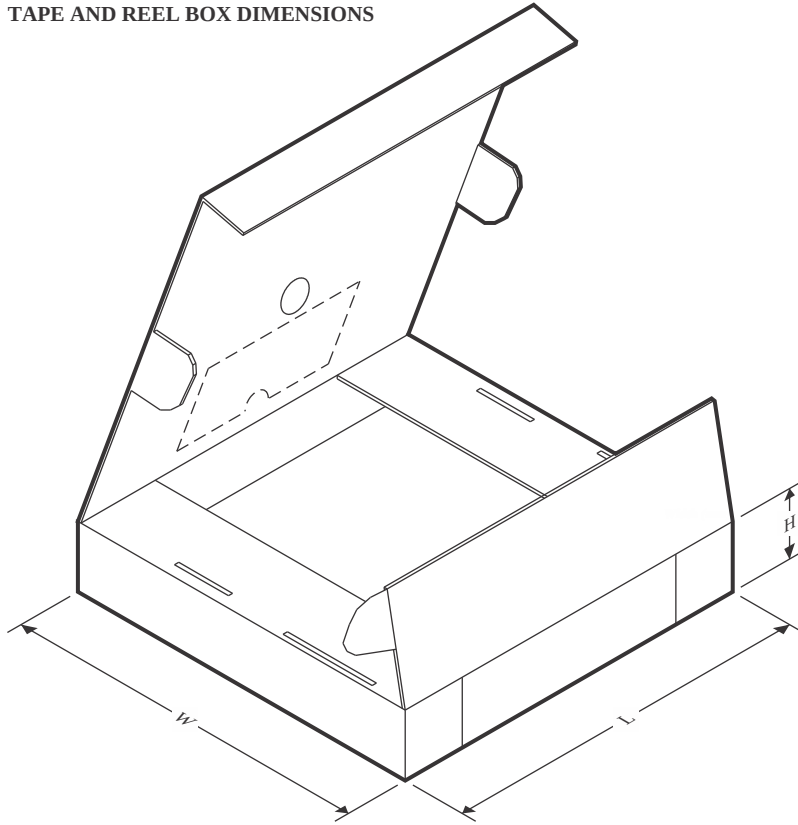
TAPE AND REEL INFORMATION



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
AMC1311BDWVR	SOIC	DWV	8	1000	330.0	16.4	12.15	6.2	3.05	16.0	16.0	Q1
AMC1311DWVR	SOIC	DWV	8	1000	330.0	16.4	12.15	6.2	3.05	16.0	16.0	Q1

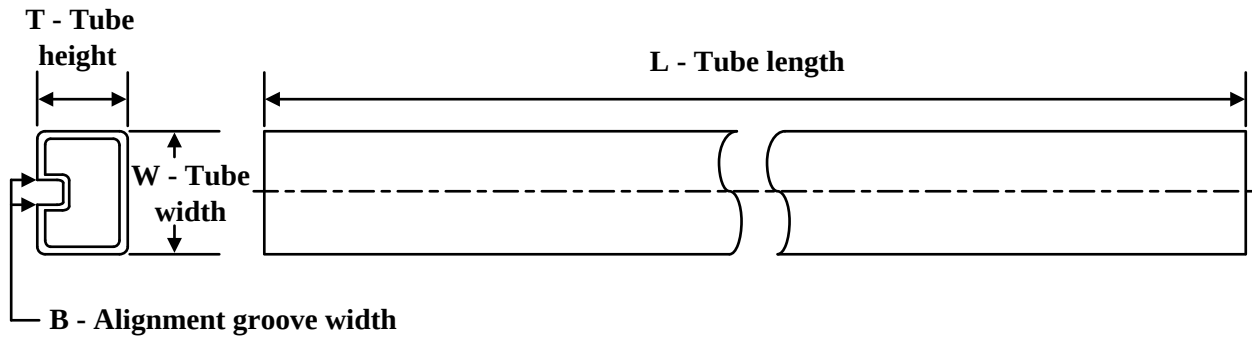
TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
AMC1311BDWVR	SOIC	DWV	8	1000	356.0	356.0	35.0
AMC1311DWVR	SOIC	DWV	8	1000	356.0	356.0	35.0

TUBE

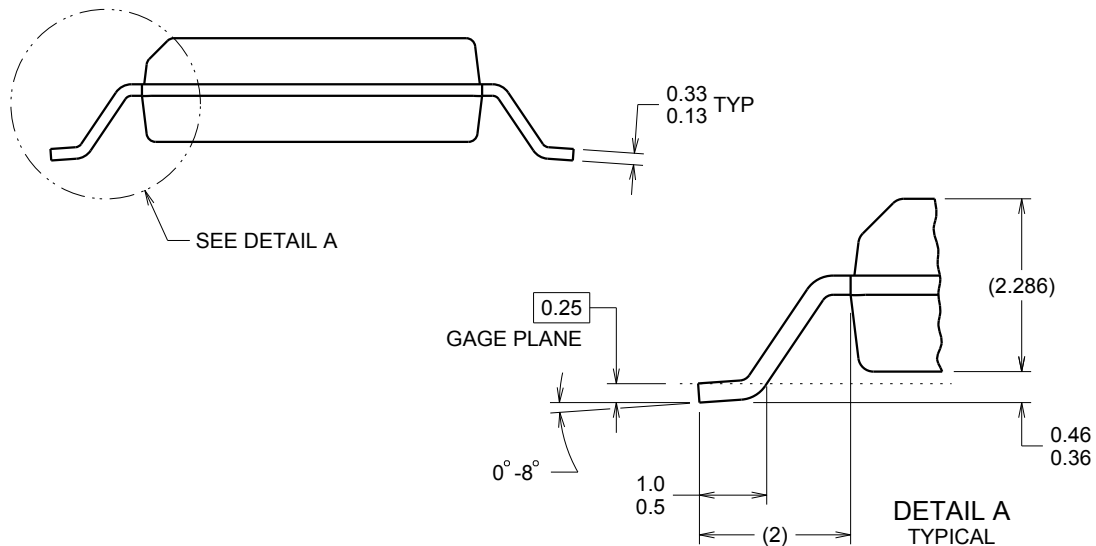


*All dimensions are nominal

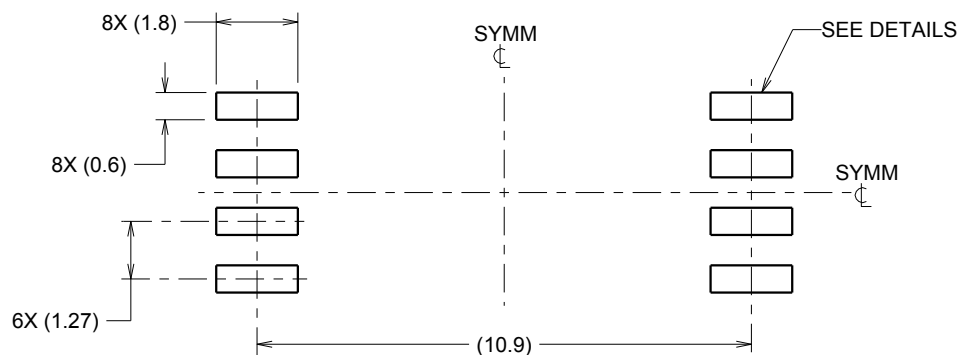
Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (μm)	B (mm)
AMC1311BDWV	DWV	SOIC	8	64	505.46	13.94	4826	6.6
AMC1311DWV	DWV	SOIC	8	64	505.46	13.94	4826	6.6

SOIC - 2.8 mm max height

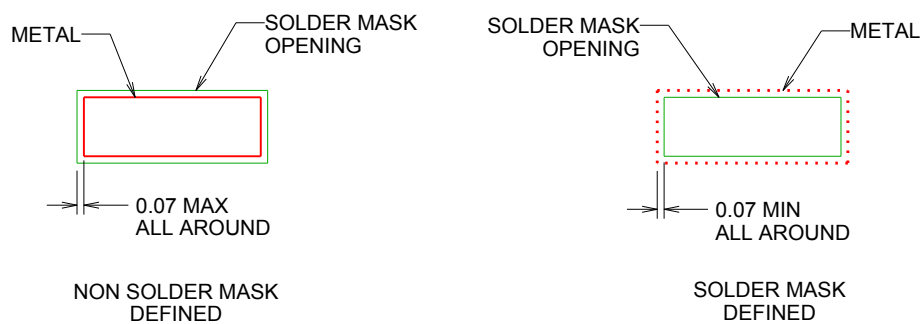
PIN 1 ID AREA
 11.5 ± 0.25 TYP
 5.95
 5.75
 NOTE 3
 1
 4
 8
 5
 6X 1.27
 2X 3.81
 8X 0.51
 0.31
 7.6
 7.4
 NOTE 4
 A
 B
 SEATING PLANE
 0.1 C
 2.8 MAX
 C
 A
 B
 S



1. All linear dimensions are in millimeters. Dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm, per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm, per side.



LAND PATTERN EXAMPLE
9.1 mm NOMINAL CLEARANCE/CREEPAGE
SCALE:6X

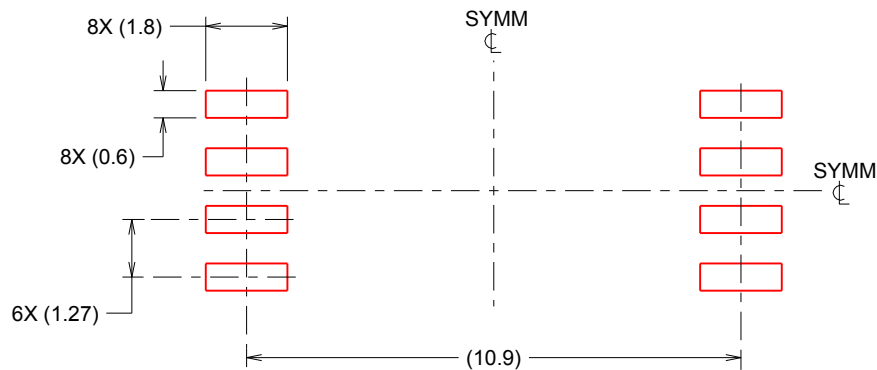


SOLDER MASK DETAILS

4218796/A 09/2013

NOTES: (continued)

5. Publication IPC-7351 may have alternate designs.
6. Solder mask tolerances between and around signal pads can vary based on board fabrication site.



SOLDER PASTE EXAMPLE
 BASED ON 0.125 mm THICK STENCIL
 SCALE:6X

4218796/A 09/2013

NOTES: (continued)

7. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
8. Board assembly site may have different recommendations for stencil design.

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